



# Maintenance Schedule

Diesel Engine

EPA

10V/12V1600Gx0S

Stationary Emergency only

Application group 3F

MSE50806/02E

| Engine model | kW/cyl.      | Application group                     |
|--------------|--------------|---------------------------------------|
| 10V1600G10S  | 46.5 kW/cyl. | 3F mains backup mode, unlimited, ICXN |
| 10V1600G20S  | 51.1 kW/cyl. | 3F mains backup mode, unlimited, ICXN |
| 12V1600G10S  | 46.8 kW/cyl. | 3F mains backup mode, unlimited, ICXN |
| 12V1600G20S  | 50.7 kW/cyl. | 3F mains backup mode, unlimited, ICXN |

*Table 1: Applicability*

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

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# 1 Maintenance Schedule

## 1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

### Definition of Rolls-Royce Solutions

Rolls-Royce Solutions refers to Rolls-Royce Power Systems AG and its subsidiaries Rolls-Royce Solutions GmbH and Rolls-Royce Solutions America Inc.

### Maintenance concept

The maintenance schedule is applicable to products certified in accordance with United States Environmental Protection Agency (U.S. EPA) and California Air Resources Board (CARB) emission regulations.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Non-observance of the maintenance instructions can result in violation of the applicable U.S. EPA and CARB emission regulations. Modification or removal of mechanical or electronic components, or the installation of additional components, as well as the execution of calibration processes which may affect the emissions characteristics of the product are prohibited by applicable U.S. EPA and/or CARB emission regulations.

Although Rolls-Royce Solutions' warranty obligations are not dependent upon the use of a specific brand of spare parts, it is recommended that emissions-relevant components be serviced, replaced or repaired using only Rolls-Royce Solutions approved components or their equivalent.

Emissions-relevant components include, but are not limited to, control units, data records, sensors, injectors, exhaust turbochargers, exhaust flaps and all components of the exhaust gas aftertreatment systems.

Use of spare parts that are not of equivalent quality to Rolls-Royce Solutions approved components may impair the effectiveness of the product's emission control systems. The maintenance, replacement, or repair of the emission control devices and systems may be performed by any repair establishment or individual of the owner's choosing.

Except for those maintenance steps designated in this Maintenance Schedule as critical emission-related maintenance, an owner or operator of the product is not required to perform the maintenance steps described in this Maintenance Schedule in order to keep the emission-related warranty valid. Rolls-Royce Solutions will not deny an emission-related warranty claim solely on the basis of an owner's or operator's failure to perform recommended additional maintenance or noncritical emission-related maintenance. However, Rolls-Royce Solutions may deny an emission-related warranty claim for the owner's or operator's failure to perform maintenance steps designated as critical emission-related maintenance or if the warranty claim arises out of failures in emission-related components that have been caused by the owner's or operator's improper maintenance or use, by accidents for which Rolls-Royce Solutions has no responsibility, or by acts of God.

The maintenance system for products from Rolls-Royce Solutions is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals, as well as the required scope of maintenance tasks, are based on operational experience.

Adherence to the specified maintenance intervals is essential to maintain product safety.

Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

The intervals according to which the maintenance tasks must be carried out are specified in terms of operating hours and time limits. The specific maintenance task is triggered when either the operating hours or time limit is first reached.

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The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by a Rolls-Royce Solutions' Service Partner. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

### **Additional notes on maintenance and preservation:**

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

## 1.2 Special conditions

| Designation                 | Value                |
|-----------------------------|----------------------|
| Load factor                 | ≤ 100%               |
| Operating hours             | Unlimited hours      |
| Time between major overhaul | 12,000 h or 18 years |

Table 2: Special conditions

### Note 01:

The engines are certified for operation with the fuels approved in the Fluids and Lubricants Specifications. For operation with a high sulfur content in the fuel, the following must be observed:

- The component TBO specified in the maintenance schedule refers to engine operation with diesel fuel with a sulfur content < 15 ppm, e.g. EN 590 or ULSD.

If a fuel with a sulfur content > 15 ppm is used, the times specified in the maintenance schedule for component TBO of the cylinder head may be shorter. For reduced TBOs, refer to the Fluids and Lubricants Specifications.

Engines with exhaust gas recirculation and/or exhaust gas aftertreatment must not be operated with excessive sulfur content in the fuel. The limit values in the Fluids and Lubricants Specifications apply.

Table 3: Special conditions

### Note 02:

If the engine is used in an UPS unit with corresponding hardware, the exchange interval for the cylinder heads is 500 h.

Table 4: Special conditions

## 1.3 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

| Interval (HRS)/limit | Limit in %           | Limit max.                                                           |
|----------------------|----------------------|----------------------------------------------------------------------|
| Operating hours      | ± 10%                | ± 500 hrs                                                            |
| Years                | ± 10%                | ± 30 days                                                            |
| Example 1            | Task every 1000 hrs  | Between 900 hrs and 1100 hrs (10% of 1000 hrs)                       |
| Example 2            | Task every 30000 hrs | Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs) |